

12.421/12.621 Lecture 12: Atmospheric Scattering and Black Body Radiation (or Why Is the Sky Blue?)

Brent Minchew

Topics covered

- Rayleigh scattering
- Mie scattering
- Planck's law

1 Rayleigh scattering

- Describes scattering from spherical particles whose radius $r \ll \lambda$ ($\lambda \equiv$ EM wavelength)
 $\Rightarrow$ Phase of EM wave $\approx$ constant
- This is simplest and very important models of EM scatter $\rightarrow$ major implications for RS
- Derivation
 - Consider small, homogeneous, isotropic, spherical particle w/ $r \ll \lambda$ (assume molecule-size particle)
 - Define incident $\mathbf{E}$ field $\mathbf{E}^i$ and let $\mathbf{E}$ be total field ($\mathbf{E} = \mathbf{E}^i + \mathbf{E}$ field caused by electric dipole)
 - Define induced dipole moment

$$\mathbf{p}_0 = \alpha_{\text{pol}} \mathbf{E}^i \quad (1)$$

where

- $\alpha_{\text{pol}} \equiv$ polarizability – an intrinsic material property that is related to the dielectric constant
 - $\hookrightarrow$ In general, α_{pol} is a 2nd-order tensor but can often be assumed to be a scalar
 - $\rightarrow$ we will take the scalar approximation
- As we discussed way back in Lecture 1, applied electric field causes dipole to oscillate in a plane that is normal to propagation vector because EM waves are transverse waves ($\mathbf{E}$, $\mathbf{H}$, and $\hat{\mathbf{k}}$ are mutually orthogonal)
- Oscillating dipole emits EM radiation $\rightarrow$ “the scattered wave”
- It can be shown (see Griffiths 2013) that dipole field at distance R from dipole can be represented as sum of

- 1) Static field $\mathbf{E}_{\text{stat}} \sim \frac{1}{R^3} \mathbf{p}$
- 2) Induction field $\mathbf{E}_{\text{ind}} \sim \frac{1}{R^2} \frac{d\mathbf{p}}{dt}$
- 3) Radiation field $\mathbf{E}_{\text{rad}} \sim \frac{1}{R} \frac{d^2\mathbf{p}}{dt^2}$

- In far-field ($R \gg \lambda$), radiation dominates, and it can be shown

$$\mathbf{E}^s = \frac{1}{c^2 R} \frac{\partial^2 \mathbf{p}}{\partial t^2} \sin \theta \quad (2)$$

where $\theta \equiv$ angle between scattered dipole moment $\mathbf{p}$ and direction of observation
(Gaussian units, as in Liou; in SI the right side carries a factor $1/(4\pi\epsilon_0)$; equivalently, replace α_{pol} by $\alpha_{\text{pol}}/(4\pi\epsilon_0)$ in what follows)

- Scattered dipole moment can be written

$$\mathbf{p} = \mathbf{p}_0 e^{j(\omega t - kR)} \quad (3)$$

- Thus, scattered $\mathbf{E}$ -field is

$$\mathbf{E}^s = -\mathbf{E}^i \frac{k^2 \alpha_{\text{pol}}}{R} \sin \theta e^{j(\omega t - kR)} \quad (4)$$

- Recall that we can represent any $\mathbf{E}$ field by orthogonal EM wave polarizations h and v (a.k.a., perpendicular and parallel), thus

$$E_h^s = -E_h^i \frac{k^2 \alpha_{\text{pol}}}{R} e^{j(\omega t - kR)} \sin \theta_1, \quad (5)$$

$$E_v^s = -E_v^i \frac{k^2 \alpha_{\text{pol}}}{R} e^{j(\omega t - kR)} \sin \theta_2, \quad (6)$$

where $\theta_1 = \frac{\pi}{2}$ by definition (see lecture notes on geometric optics)

and $\theta_2 = \frac{\pi}{2} - \Theta$ where $\Theta \equiv$ scattering angle (angle between incident & scattered waves)

- We are interested in the intensity I , defined as power per unit area, which is related to $\mathbf{E}$ -field magnitude as

$$I_p \sim |E_p^s|^2 \quad (7)$$

- Therefore

$$I_h = I_{0h} \frac{k^4 \alpha_{\text{pol}}^2}{R^2}, \quad (8)$$

$$I_v = I_{0v} \frac{k^4 \alpha_{\text{pol}}^2}{R^2} \cos^2 \Theta \quad (9)$$

$$\Rightarrow I = I_h + I_v = (I_{0h} + I_{0v} \cos^2 \Theta) \frac{k^4 \alpha_{\text{pol}}^2}{R^2}$$

- Let us consider case of scattering from Earth's atmosphere $\rightarrow$ sunlight is unpolarized $\Rightarrow I_{0h} = I_{0v} = \frac{I_0}{2}$
where $I_0 \equiv$ incident intensity

$$\Rightarrow \boxed{I = I_0 \frac{\alpha_{\text{pol}}^2}{R^2} k^4 \frac{1 + \cos^2 \Theta}{2} = I_0 \frac{\alpha_{\text{pol}}^2}{R^2} \left(\frac{\omega}{c}\right)^4 \frac{1 + \cos^2 \Theta}{2}} \quad (10)$$

- This is the formula originally derived by Rayleigh in 1871

– Take-homes

- Scattered intensity $\sim R^{-2}$ (due to spreading losses)

• Scattered intensity $\sim \omega^4$ (equiv. $\frac{1}{\lambda^4}$)
 $\Rightarrow$ higher frequency radiation scatters more strongly

- Scattered intensity $\sim \alpha_{\text{pol}}^2 \Rightarrow$ relatable to dielectric constant ε of atmosphere; can be shown that:

$$\alpha_{\text{pol}} \sim \left(\frac{\varepsilon - 1}{\varepsilon + 2} \right) \quad (\text{“Lorentz–Lorenz formula”}) \quad (11)$$

$\Rightarrow$ Intensity of scattered light increases w/ ε

Moisture influences scattered intensity even in clear skies, but mainly by swelling aerosol particles (hygroscopic growth); water vapor itself changes molecular (Rayleigh) scattering only slightly (particulates will play an important role, as we will see in a bit)

- Polarization of light: dependence on scattering angle Θ only comes into vertically polarized term; $I_v \sim \cos^2 \Theta \Rightarrow \Theta = \frac{\pi}{2}$ has only horizontally polarized light
 $\Rightarrow$ atmosphere acts as a polarizing filter, something that photographers who use polarizing filters will know very well

- An extension of the Rayleigh scattering model we just derived that accounts for particle size r ($r \ll \lambda$) gives

$$I = I_0 \left(\frac{\omega}{c} \right)^4 \frac{r^6}{R^2} \frac{1 + \cos^2 \Theta}{2} \left(\frac{n^2 - 1}{n^2 + 2} \right)^2 \quad (12)$$

where $n^2 = \varepsilon$ ($n \equiv$ index of refraction)

$\Rightarrow$ Scattered intensity scales as $I \sim r^6$ (for $r \ll \lambda$)

- What about when $r \gtrsim \lambda$?

2 Mie scattering

- Key approximation in Rayleigh scattering is that $r \ll \lambda$ meaning we can ignore the $\mathbf{E}$ field inside the particle and approximate scattered radiation as having same phase $\rightarrow$ works for clear atmosphere
- As $r \rightarrow \lambda$, we have to allow EM phase to vary over surface and for particle to contain $\mathbf{E}$ -fields $\Rightarrow$ interference

$\rightarrow$ generally the case when particles or aerosols are present

- Formal derivation of Mie scattering beyond the scope of this course
- It can be shown that the scattered intensity of a spherical particle is, for small kr ,

$$I = I_0 \left(\frac{\omega}{c} \right)^4 \frac{r^6}{R^2} \frac{1 + \cos^2 \Theta}{2} \left(\frac{n^2 - 1}{n^2 + 2} \right)^2 (1 + c_1 k^2 r^2 + c_2 k^4 r^4 + \dots) \quad (13)$$

where

$$c_1 = \frac{6}{5} \left(\frac{n^2 - 2}{n^2 + 2} \right), \quad (14)$$

$$c_2 = \frac{3}{175} \frac{n^6 + 41n^4 - 284n^2 + 284}{(n^2 + 2)^2} + \frac{1}{900} \left(\frac{n^2 + 2}{2n^2 + 3} \right)^2 [15 + (2n^2 + 3)^2] \quad (15)$$

(The coefficients c_1, c_2 are those of the scattered power integrated over all directions; at a fixed scattering angle the correction differs, e.g., for $n = 1.33$ and $kr = 0.3$ the exact ratio to the Rayleigh intensity is 1.033 forward, 0.992 at 90° , and 0.954 backward, versus $1 + c_1 k^2 r^2 = 0.993$.)

- It's clear from the above that when $r \ll \lambda$ ($kr \ll 1$), the Mie scattering expression above reduces to the Rayleigh approximation.
- For $r \gtrsim \lambda$, interference is present causing I to oscillate in frequency space. Conceptually (Figure 1)

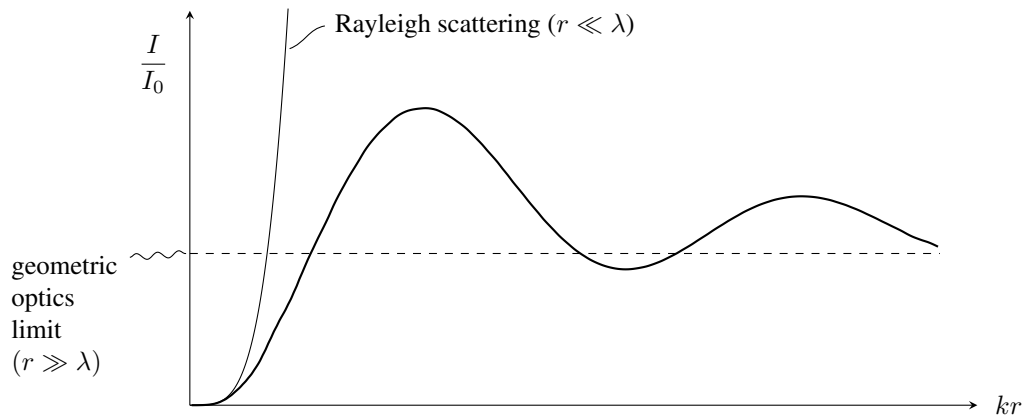


Figure 1: Scattered intensity (normalized by the incident intensity) versus kr : Rayleigh scattering for $r \ll \lambda$, oscillations due to interference for $r \gtrsim \lambda$, and the geometric optics limit for $r \gg \lambda$. Curve computed from the exact Mie solution for nonabsorbing spheres with $n = 1.33$ (averaged over a narrow range of sizes), as the scattered power integrated over all directions divided by $I_0 \pi r^2$.

3 Solar radiation at the top of the atmosphere

- Sun is well-represented as a black body (an idealized structure that absorbs all incident EM radiation regardless of frequency and incidence angle)
- Visualize black-body
Large cavity w/ opaque walls and a small hole (Figure 2)

3.1 Planck's law

- To derive theory for cavity radiation, Planck assumed atoms in wall were tiny oscillators (just like we did in Lecture 1) each w/ characteristic oscillation frequency
- Oscillators emit energy into and absorb energy from cavity
- Planck made 2 assumptions about oscillators

1. Energy given by

$$\mathcal{E} = \hbar\omega \left(n_{\text{vib}} + \frac{1}{2} \right) \quad (16)$$

where n_{vib} = vibrational quantum number as we saw last lecture

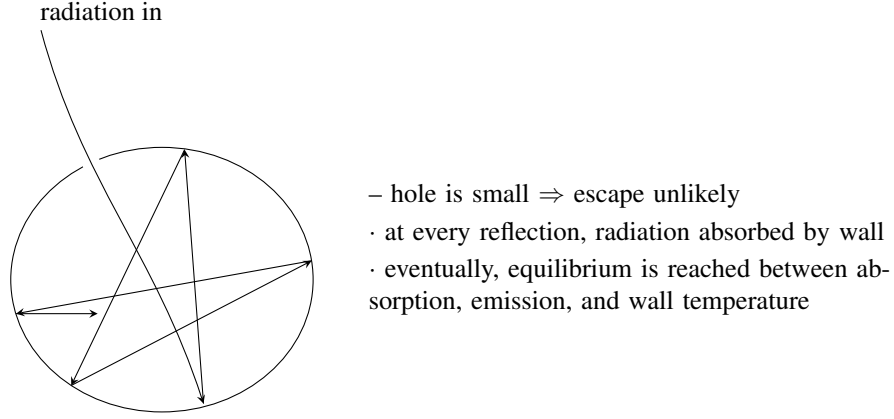


Figure 2: A black body visualized as a large cavity with opaque walls and a small hole: radiation entering the cavity is repeatedly reflected and absorbed by the walls.

2. Energy is quantized

$$\Delta\mathcal{E} = \hbar\omega \quad (17)$$

- Determining emitted energy requires knowing total number of oscillators w/ freq ω for all possible states
- From Boltzmann statistics, number of oscillators in a state with energy $\mathcal{E}_{\text{ex}}$ above the ground state is

$$N = N_0 e^{-\mathcal{E}_{\text{ex}}/k_B T} \quad (18)$$

$k_B \equiv$ Boltzmann's const,

$T \equiv$ absolute temperature.

- Energy is quantized, thus $\mathcal{E}_{\text{ex}}$ takes on discrete values

$$\mathcal{E}_{\text{ex}} = 0, \hbar\omega, 2\hbar\omega, \dots$$

- Therefore, total number of oscillators w/ freq. ω

$$\begin{aligned} N_\omega &= N_0 + N_0 e^{-\hbar\omega/k_B T} + N_0 e^{-2\hbar\omega/k_B T} + \dots \\ &\simeq \frac{N_0}{1 - e^{-\hbar\omega/k_B T}} \end{aligned} \quad (19)$$

- Total energy of oscillators comes from multiplying each term by appropriate energy level

$$\begin{aligned} \mathcal{E}_{\text{tot}} &= 0 \cdot N_0 + \hbar\omega N_0 e^{-\hbar\omega/k_B T} + 2\hbar\omega N_0 e^{-2\hbar\omega/k_B T} + \dots \\ &\simeq \frac{\hbar\omega N_0 e^{-\hbar\omega/k_B T}}{(1 - e^{-\hbar\omega/k_B T})^2} \end{aligned} \quad (20)$$

- Average energy per oscillator is

$$\frac{\mathcal{E}_{\text{tot}}}{N_\omega} = \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1} \quad (21)$$

- Now define monochromatic energy density u_f (energy per volume per frequency)

$$u_f = A \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1} \quad (22)$$

where

$$A = \frac{8\pi f^2}{c^3} \quad \text{with } f = \frac{\omega}{2\pi}, \quad c \equiv \text{speed of light}$$

The value of A ensures that u_f tends to the Rayleigh–Jeans law at low frequencies

- Finally, define spectral radiance (power per unit solid angle per unit projected area per unit frequency $\Rightarrow$ units $\text{W sr}^{-1} \text{m}^{-2} \text{Hz}^{-1}$)

$$B_f = \frac{u_f c}{4\pi} = \frac{2hf^3}{c^2} \left[\exp \left\{ \frac{hf}{k_B T} \right\} - 1 \right]^{-1} \quad (23)$$

which can be written in terms of wavelength as

$$B_\lambda = -B_f \frac{df}{d\lambda} = \frac{2hc^2}{\lambda^5} \left[\exp \left\{ \frac{hc}{\lambda k_B T} \right\} - 1 \right]^{-1} \quad (24)$$

where

$$h = 2\pi\hbar \equiv \text{Planck's constant}$$

- Take-homes

- Strong frequency (and wavelength) dependence
 - $\hookrightarrow$ peak spectral radiance (of B_λ) at wavelength

$$\lambda_{\max} = \frac{b_W}{T} \quad (25)$$

Known as Wien's displacement

where $b_W = 2.898 \times 10^{-3} \text{ m}\cdot\text{K} \equiv \text{Wien's displacement const}$

- $\hookrightarrow$ rapid drop off in radiance for $\lambda < \lambda_{\max}$ because it is hard for photons with energy $\gg k_B T$ (mean or characteristic energy) to be emitted
- $\hookrightarrow$ long tail at $\lambda > \lambda_{\max}$ due to Boltzmann statistics
- Strong T dependence $\rightarrow$ influences both total radiated energy and distribution
- For Sun ($T \sim 5800 \text{ K}$ at surface), highest radiance in blue spectrum
 - $\hookrightarrow$ Recalling Rayleigh scattering intensity $\sim \omega^4$ we get blue skies (complemented by absorption by ozone in atmosphere)
 - $\hookrightarrow$ Why not a violet sky? $\lambda^{-4} B_\lambda$ ($T = 5800 \text{ K}$) keeps rising to shorter λ (peak at $0.276 \mu\text{m}$, in the UV; ozone absorbs the solar UV below $\sim 0.3 \mu\text{m}$), but observed sunlight holds much less energy in the violet than in the blue, and the eye is much less sensitive to violet

3.2 Stefan–Boltzmann law

- Finally, in RS it is sometimes the case that we only measure the total radiant intensity

$$\begin{aligned}
 L(T) &= \int_0^\infty B_\lambda(T) d\lambda \\
 &= \frac{2k_B^4 T^4}{h^3 c^2} \underbrace{\int_0^\infty \frac{z^3}{e^z - 1} dz}_{=\pi^4/15}
 \end{aligned} \tag{26}$$

where $z = \frac{hc}{\lambda k_B T}$

$$\Rightarrow L(T) = bT^4 \quad \text{where } b = \frac{2\pi^4 k_B^4}{15c^2 h^3} = \text{const}$$

- Since black body radiation is isotropic, intensity

$$I = \pi L = \sigma_{\text{SB}} T^4, \quad \sigma_{\text{SB}} = 5.67 \times 10^{-8} \left[\frac{\text{W}}{\text{m}^2 \text{K}^4} \right] \equiv \text{Stefan–Boltzmann const} \tag{27}$$

Variables

Units are SI. The Rayleigh derivation (Section 1) follows Liou's Gaussian form, in which α_{pol} has the dimension of volume; fields and dipole moments are listed below with their SI units, and in SI the scattered-field formulas carry the factor $1/(4\pi\epsilon_0)$ (equivalently, $\alpha_{\text{pol}} \rightarrow \alpha_{\text{pol}}/(4\pi\epsilon_0)$, with α_{pol} in $\text{C m}^2/\text{V}$). Values are given for universal constants.

Symbol	Description	Units	Value
<i>Constants</i>			
c	speed of light in a vacuum	m/s	2.998×10^8
h	Planck constant	J s	6.626×10^{-34}
$\hbar$	reduced Planck constant, $h/2\pi$	J s	1.055×10^{-34}
k_B	Boltzmann constant	J/K	1.381×10^{-23}
ϵ_0	permittivity of free space	F/m	8.854×10^{-12}
σ_{SB}	Stefan–Boltzmann constant	$\text{W m}^{-2}\text{K}^{-4}$	5.67×10^{-8}
b_W	Wien's displacement constant	m K	2.898×10^{-3}
b	constant in $L = bT^4$, σ_{SB}/π	$\text{W m}^{-2} \text{sr}^{-1} \text{K}^{-4}$	1.805×10^{-8}
<i>Rayleigh and Mie scattering</i>			
r	particle radius	m	$\ll \lambda$ (Rayleigh)
λ	EM wavelength	m	–
ω	angular frequency	rad/s	–
k	wavenumber in the surrounding medium (air), $\omega/u_p \approx \omega/c$ (D163)	rad/m	$\approx 2\pi/\lambda$
R	distance from the dipole (scatterer)	m	far field: $R \gg \lambda$
t	time	s	–
$\mathbf{E}^i$	incident electric field	V/m	–
$\mathbf{E}$	total electric field, $\mathbf{E}^i$ plus the dipole's field	V/m	–
$\mathbf{E}^s$	field scattered by the dipole	V/m	–
$\mathbf{E}_{\text{stat}}$	static field of a dipole	V/m	$\sim R^{-3}$
$\mathbf{E}_{\text{ind}}$	induction field of a dipole	V/m	$\sim R^{-2}$
$\mathbf{E}_{\text{rad}}$	radiation field of a dipole	V/m	$\sim R^{-1}$
E_h^s, E_v^s	h - and v -polarized scattered field (perpendicular and parallel to the scattering plane)	V/m	–
E_h^i, E_v^i, E_p^i	h -, v - and p -polarized incident field	V/m	–
$\mathbf{H}$	magnetic field	A/m	–
$\hat{\mathbf{k}}$	propagation unit vector	dimensionless	–
$\mathbf{p}, \mathbf{p}_0$	(scattered) dipole moment; its amplitude	C m	$\mathbf{p}_0 = \alpha_{\text{pol}} \mathbf{E}^i$
α_{pol}	polarizability	cm^3 (Gaussian)	$r^3(n^2 - 1)/(n^2 + 2)$ (sphere)
θ	angle between $\mathbf{p}$ and the direction of observation	rad	–
θ_1, θ_2	θ for the h and v components	rad	$\pi/2; \pi/2 - \Theta$
Θ	scattering angle	rad	$0 - \pi$
I	intensity (power per unit area)	W/m^2	–
I_h, I_v, I_p	h -, v - and p -polarized scattered intensity	W/m^2	–

Symbol	Description	Units	Value
I_0, I_{0h}, I_{0v}	incident intensity; its h and v parts	W/m^2	$I_{0h} = I_{0v} = I_0/2$ (unpolarized)
p	(subscript) polarization, h or v	–	–
ε	dielectric constant	dimensionless	–
n	index of refraction	dimensionless	$\sqrt{\varepsilon}$
c_1, c_2	coefficients of the Mie correction series	dimensionless	Eqs. (14)–(15)
<i>Black body radiation</i>			
T	(absolute) temperature	K	Sun: ~ 5800
f	frequency, $\omega/2\pi$	Hz	–
$\mathcal{E}$	energy of an oscillator	J	$\hbar\omega(n_{\text{vib}} + \frac{1}{2})$
$\mathcal{E}_{\text{tot}}$	total energy of the oscillators with frequency ω	J	Eq. (20)
$\Delta\mathcal{E}$	quantum of energy emitted or absorbed	J	$\hbar\omega$
$\mathcal{E}_{\text{ex}}$	energy of a state above the ground state	J	$0, \hbar\omega, 2\hbar\omega, \dots$
n_{vib}	vibrational quantum number	dimensionless	$0, 1, 2, \dots$
N	number of oscillators with energy $\mathcal{E}_{\text{ex}}$ above the ground state	dimensionless	–
N_0	number of oscillators in the ground state	dimensionless	–
N_ω	total number of oscillators with frequency ω	dimensionless	–
u_f	monochromatic energy density (per unit frequency)	$\text{J m}^{-3}\text{Hz}^{-1}$	–
A	number of modes per unit volume per unit frequency	$\text{m}^{-3}\text{Hz}^{-1}$	$8\pi f^2/c^3$
B_f	spectral radiance (per unit frequency)	$\text{W m}^{-2} \text{sr}^{-1} \text{Hz}^{-1}$	–
B_λ	spectral radiance (per unit wavelength)	$\text{W m}^{-2} \text{sr}^{-1} \text{m}^{-1}$	–
λ_{max}	wavelength of the peak of B_λ	m	b_W/T
L	total radiance (“total radiant intensity”)	$\text{W m}^{-2}\text{sr}^{-1}$	bT^4
z	integration variable, $hc/(\lambda k_B T)$	dimensionless	–
I	intensity (flux density) emitted by a black body	W/m^2	$\pi L = \sigma_{\text{SB}} T^4$